11.22.2.3. BDX Initiator

The node that initiates a bulk data transfer. The Initiator of a data transfer can either be the Sender (for "upload", which starts with a `SendInit`) or the Receiver (for "download", which starts with a `ReceiveInit`).

11.22.2.4. BDX Responder

The node that responds to the initiator by either accepting or rejecting the proposed bulk data transfer and choosing parameters of the transfer compatible with those proposed by the Initiator. It can also either be the Sender (for "download", which is when the Responder receives a `ReceiveInit`) or the Receiver (for "upload", which is when the Responder receives a `SendInit`).

11.22.2.5. BDX Synchronous / Asynchronous Modes

Bulk data transfers can operate in synchronous ("driven") or asynchronous mode. When operating in synchronous mode, one party (the Driver) is responsible for controlling the rate at which the transfer proceeds, and each message in the bulk data transfer protocol SHALL be acknowledged before the next message will be sent.

In asynchronous mode, there is no driver and successive messages are freely sent without waiting for `BlockAck` responses from the Receiver. In asynchronous mode, flow control is provided by the underlying transport (e.g. Matter over TCP).

NOTE Support for the asynchronous mode is provisional.

11.22.2.6. BDX Driver

The node (either Sender or Receiver) that controls the rate at which a synchronous data transfer proceeds by sending `Block` or `BlockQuery` messages to advance the transfer. This facility allows for intermittently connected devices operating (e.g. due to battery constraints) to complete a bulk data transfer without requiring them to stay active continuously during the transfer. In every synchronous bulk data transfer, exactly one device acts as Driver, and the transfer consists of a series of request/responses, each one initiated by the Driver.

BDX utilizes features of the underlying message layer to provide reliability and at-most-once delivery semantics. If the transport fails to deliver the message within the specified parameters, the BDX session SHOULD be aborted. For synchronous transfers, if the Driver fails to receive the response to any given request that is not received within a particular application-determined time, it SHOULD abort the session.

11.22.2.7. BDX Follower

The node (either Sender or Receiver) that "follows" the driver in the protocol flow. The protocol defines the followers as devices that can never be sleepy. Follower either receives a `BlockQuery` to send the data upon request from the driver or receives a new `Block` message and acknowledges it with a `BlockAck` message (in synchronous mode).

